Solving Problems

Effectively collaborating with Ai

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The Jackson Laboratory

Day 3 Data Literacy & Examining the Machine

How does data deceive us?
How do we audit machines and **ourselves?**

Workshop Overview: *Why do smart people make predictable mistakes?*

Day 1- What is Knowledge?

- Discussion: the crisis of authority and need for metacognition
- What is knowledge?
- Kahneman systems
- Intro to 4 frameworks

Day 2 – What is Reasoning?

- Discussion: CRT & Paper
- Structure of an Argument
- Paul-Elder Framework
- Intellectual Autopsy
- Polya’s 4 steps

Day 3 – Data Literacy

- Discussion: Dataset
- Epistemology of Data
- Wild & Pfannkuch PPDAC Cycle
- Synthesis of the four frameworks
- Turing Audit

The Calibration Problem: How do we know when to trust our own conclusions?

The Four Frameworks

Kahneman

System 1 - *FAST*
- Uses heuristics

System 2 – *SLOW*
- Deliberate, and analytical
- Cognitively demanding

Day 1

Paul-Elder

1. Purpose
2. Question at Issue
3. Information
4. Interpretation & Inference
5. Concepts
6. Assumptions
7. Implications & Consequences
8. Point of View

Polya

1. Understand Problem
2. Devise a Plan
3. Execute Plan
4. Iteration & Reflection

Day 2

Wild & Pfannkuch

1. Problem
2. Plan
3. Data
4. Analysis
5. Conclusion

Day 3

The Four Frameworks

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Calibration failures these model's address:

We are unaware that fast thinking is operating at all!

We argue from hidden assumptions that we never examine!

We treat complex problems as unsolvable wholes!

We trust data that encodes errors that we cannot see!

Thinker's Log

Entry 3

Thinker's Log Entry #3

Prompt 1: What did I believe confidently this week that turned out to be wrong or uncertain?

Prompt 2: What evidence would change your most constant belief?

Prompt 3: Describe a time when a number or statistic changed your mind. Was the change justified?

Yesterday's Homework

Discussion

Discussion

- Name one assumption you made recently that you hadn't examined before today.
- (optional) Re-read one of your own recent abstracts, grant aims or other and map it to the 8 elements. Identify the weakest element.
- **Mini-Case Study. Day 2:** Hospital considering adoption of an Ai diagnostic system (introduced in class). Audit the reasoning for your recommendation yesterday.
- **Personal Heuristic Protocol, Day 2:** note your suspected most-common reasoning failure modes

The Assumption Auction

Activity #1

The Assumption Auction

1. People who speak confidently usually know what they're talking about.
2. More data automatically leads to better decisions.
3. Most disagreements in teams come from poor communication.
4. If everyone follows the protocol, the outcome will be fair.
5. Experts are less biased than non-experts.
6. Students learn best when information is presented clearly.
7. A successful project is usually the result of good planning.
8. When a result is statistically significant, it is probably important.

The Assumption Auction

Join by Web **PollEv.com/daniellepresgraves863**

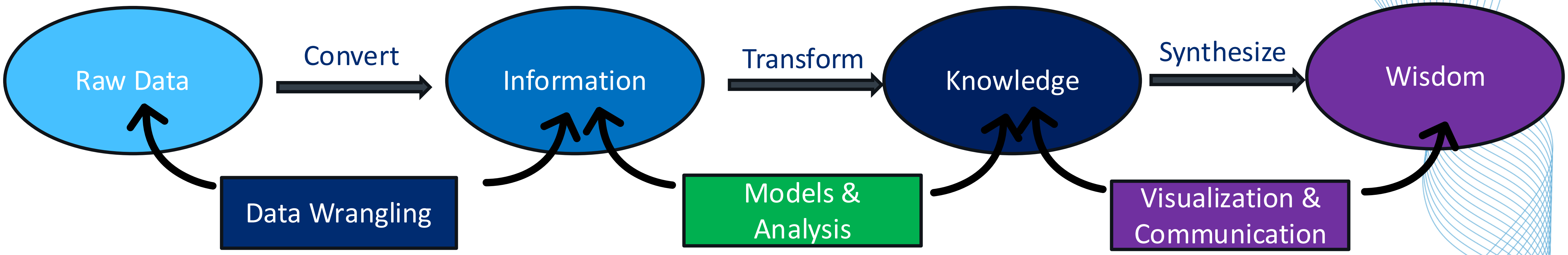
Epistemology of Data

Lecture

Epistemology of Data: Data is a measurement of reality, not reality itself.

200,000

Epistemology of Data: Data is NOT reality, it is a representation



Reality → Observation → Measurement → Recording → Aggregation → Number

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200,000

https://www.npr.org/sections/coronavirus-live-updates/2020/09/22/911934489/enormous-and-tragic-u-s-has-lost-more-than-200-000-people-to-covid-19

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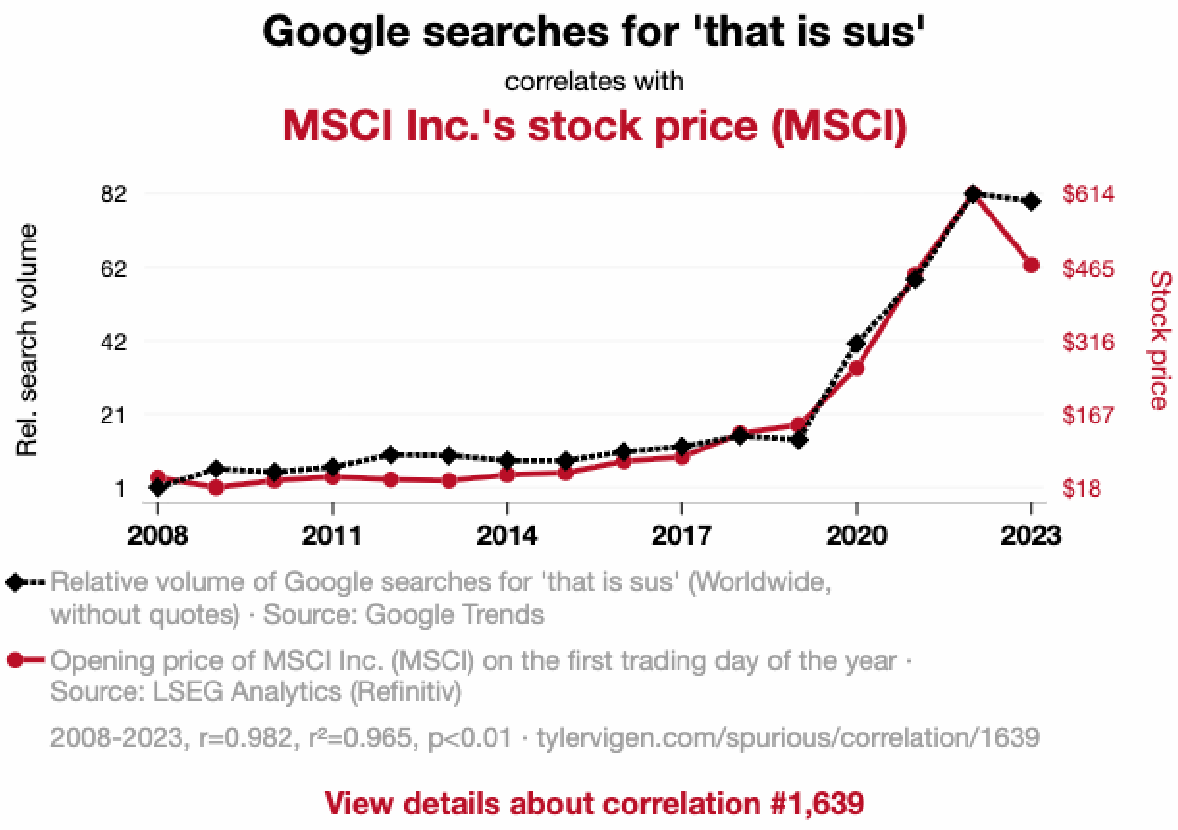
Epistemology of Data: The Sources of Data Error

1. **Measurement Error** – the instrument is imprecise or invalid

<u>Examples:</u>

- Using proxies
- Observer effect



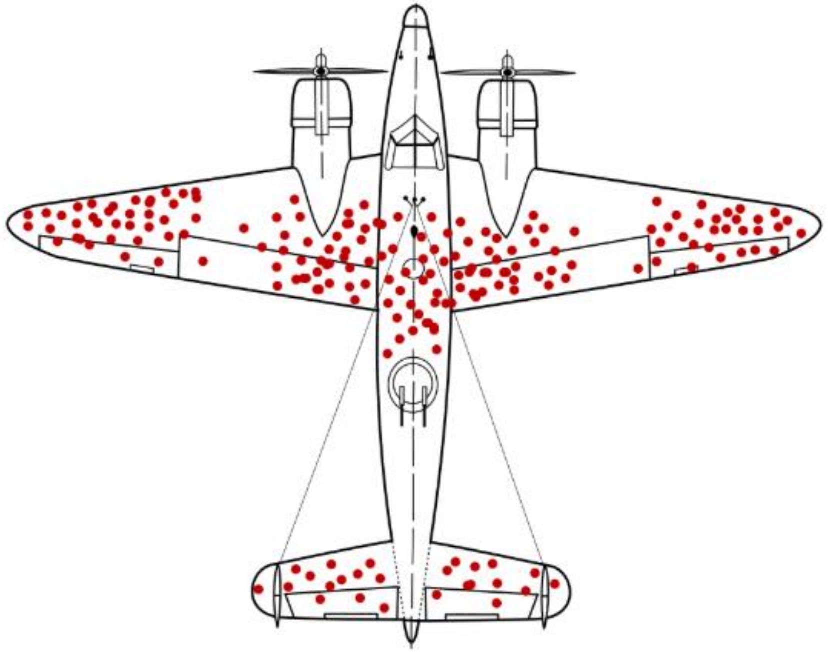
https://www.tylervigen.com/spurious-correlations

Epistemology of Data: The Sources of Data Error

1. **Measurement Error** – the instrument is imprecise or invalid

2. **Sampling Error** - observations do not represent the population of interest

- not random sampling, but rather systematic sampling bias
- Examples:
  - WEIRD bias in research
  - Survivorship bias
  - Temporal Sampling



https://en.wikipedia.org/wiki/Survivorship_bias

Epistemology of Data: The Sources of Data Error

1. **Measurement Error** – The instrument is imprecise or invalid
2. **Sampling Error** - Observations do not represent the population of interest
3. **Aggregation Error** - Combining data obscures the variation that matters

Patient Group	Volume	Hospital Death Rate	Competitor Death Rate
Routine Cases	High (e.g., 1500)	Excellent (3.5%)	Poor (4.5%)
Complex Cases	Low (e.g., 200)	Poor (40.0%)	Excellent (15.0%)

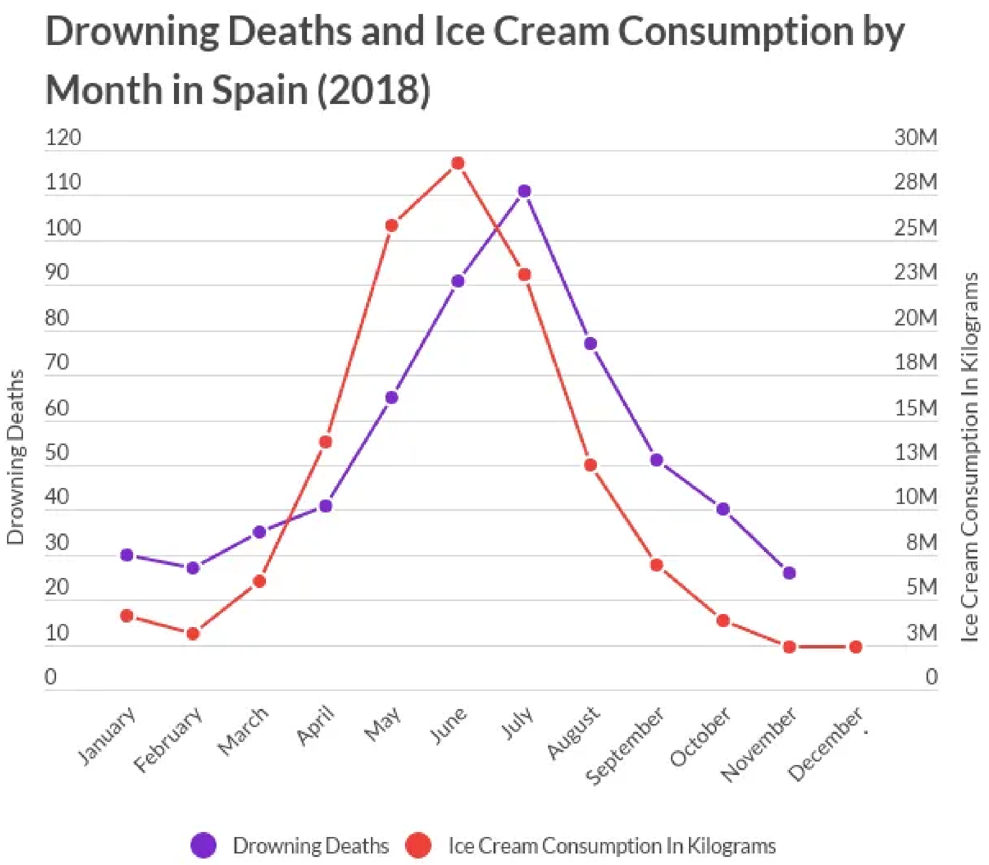
Epistemology of Data: The Sources of Data Error

1. **Measurement Error** – The instrument is imprecise or invalid
2. **Sampling Error** - Observations do not represent the population of interest
3. **Aggregation Error** - Combining data obscures the variation that matters

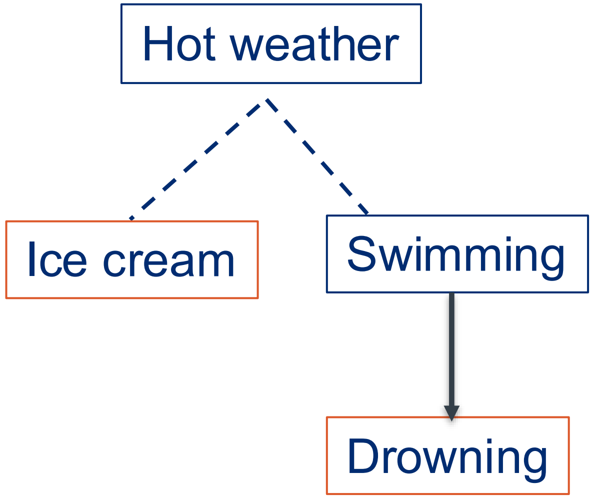
Why do these errors matter?

Because they make us think something causes something else when it doesn't!

Correlation does not imply Causation, except sometimes it does



Statista (2020)



Correlation is evidence; not an explanation

https://andreasrmadsen.medium.com/a-story-of-ice-cream-drowning-and-causal-modelling-fff3967f7671

Correlation does not imply Causation, except sometimes it does

Association ≠ Prediction ≠ Causation

Association	What moves together?	Statistical Significance (Science)
Prediction	What can I forecast?	Machine Learning (Ai)
Causation	What happens if I **intervene?**	Real-world (clinical) decisions

Epistemology of Data: Biography of a Number

Four questions that constitute a useful heuristic

Question 1: *What was actually measured?*

Question 2: *Who was measured, and who was not?*

Question 3: *How was "it" defined and did that definition stay stable?*

Question 4: *What does this number <u>not</u> tell us?*

Epistemology of Data: Data is a measurement of reality, not reality itself.

200,000

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The Wild & Pfannkuch PPDAC Cycle

Lecture

Statistical Thinking as cognitive practice

WHAT IS STATISTICAL THINKING?

Chris Wild and Maxine Pfannkuch,
Department of Statistics and Mathematics Education Unit,
The University of Auckland, New Zealand

This paper begins an exploration of "statistical thinking". We take as our starting point the conception that the ultimate goal of (applied) statistical investigation is learning in the context sphere of the problem and that "statistical thinking" is the interplay between the statistical and the contextual. This macro-level discussion of statistical thinking draws upon interviews with statistics students and professional statisticians. We concentrate on broad aspects of thinking skills required in applied statistics, indicate peculiarly statistical aspects, and make recommendations for further work. Along the way we discuss contributions of the quality movement to statistical thinking.

- Beyond just the official hypothetico-deductive framework
- What do expert statisticians ***actually do*** when they reason well?
- Gap between novice and expert is not mathematical sophistication
- Does the reasoner understand the entire cycle from problem to conclusion and do they iterate across the cycle
- PPDAC is the result – it is an **empirical description of good statistical thinking**

Statistical Thinking in Empirical Enquiry

C.J. Wild and M. Pfannkuch

Department of Statistics, University of Auckland, Private Bag 92019, Auckland, New Zealand

Summary

This paper discusses the thought processes involved in statistical problem solving in the broad sense from problem formulation to conclusions. It draws on the literature and in-depth interviews with statistics students and practising statisticians aimed at uncovering their statistical reasoning processes. From these interviews, a four-dimensional framework has been identified for statistical thinking in empirical enquiry. It includes an investigative cycle, an interrogative cycle, types of thinking and dispositions. We have begun to characterise these processes through models that can be used as a basis for thinking tools or frameworks for the enhancement of problem-solving. Tools of this form would complement the mathematical models used in analysis and address areas of the process of statistical investigation that the mathematical models do not, particularly areas requiring the synthesis of problem-contextual and statistical understanding. The central element of published definitions of statistical thinking is "variation". We further discuss the role of variation in the statistical conception of real-world problems, including the search for causes.

Key words: Causation; Empirical investigation; Statistical thinking framework; Statisticians' experiences; Students' experiences; Thinking tools; Variation.

1 Introduction

"We all depend on models to interpret our everyday experiences. We interpret what we see in terms of mental models constructed on past experience and education. They are constructs that we use to understand the pattern of our experiences." David Bartholomew (1995).

"All models are wrong, but some are useful" George Box

This paper abounds with models. We hope that some are useful!

This paper had its genesis in a clash of cultures. Chris Wild is a statistician. Like many other statisticians, he has made impassioned pleas for a wider view of statistics in which students learn "to think statistically" (Wild, 1994). Maxine Pfannkuch is a mathematics educator whose primary research interests are now in statistics education. Conception occurred when Maxine asked "What *is* statistical thinking?" It is not a question a statistician would ask. Statistical thinking is the touchstone at the core of the statistician's art. But, after a few vague generalities, Chris was reduced to stuttering.

The desire to imbue students with "statistical thinking" has led to the recent upsurge of interest in incorporating real investigations into statistics education. However, rather than being a precisely understood idea or set of ideas, the term "statistical thinking" is more like a mantra that evokes things understood at a vague, intuitive level, but largely unexamined. Statistical thinking is the statistical incarnation of "common sense". "We know it when we see it", or perhaps more truthfully,

Problem: *What are we actually trying to find out?*

1. **Real-world problem** → the focus that matters
    - What causes Alzheimer's?

2. **Statistical question** → tractable, measurable version of real-world problem
    - Does Aβ*56 correlate with memory impairment in transgenic mice?

Reality → Observation → Measurement → Recording → Aggregation → Number

Problem: *What are we actually trying to find out?*

Decisions get made at this stage and they are often not documented!

1. **Scope**
   - What counts as an instance? What time period?
   - Who is included in the target population?
2. **Operationalization**
   - How will abstract concepts be translated into measurable variables?
3. **Purpose**
   - Is this exploratory or are we testing a hypothesis?

Reality → Observation → Measurement → Recording → Aggregation → Number

PPDAC – Plan

Plan: *How to collect appropriate data that can answer our question?*

Measurement decisions – and their resultant errors – live here. You cannot rescue a flawed plan even with beautiful statistics.

1. **Study Design**
   - What type of evidence can be produced with this design?
   - Experimental (RCT) versus observational?
2. **Measurement Instrument**
   - What tool and what are its specificity, sensitivity, bias, and limitations?
   - Is this exploratory or are we testing a hypothesis?
3. **Sample Design**
   - Who, how, and what? Beware of bias in sampling

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PPDAC – Data

Data: *Is what we collected what we planned to collect?*

Data collection, cleaning, and preparation.

1. **Data Quality problems**
   - Missing values, inconsistencies across sites/times, data entry errors
2. **Protocol deviation**
   - What was done compared to what the plan said they would do
3. **Any gaps between planned and actual data**
   - Delta between who you recruited and who you intended to recruit
   - Is this gap systematic in ways that influence generalizability?

Reality → Observation → Measurement → Recording → Aggregation → Number

Analysis: *Are we drawing the right conclusions from the data we have?*

Statistics stage. Errors typically originate as framing errors (wrong question, or inappropriate method)

1. **Confounding variables**
2. **Multiple comparisons without adjusting α**
3. **Overfitting**
4. **Inappropriate conclusions**

Reality → Observation → Measurement → Recording → Aggregation → Number

Conclusion: *What are we entitled to claim (and not entitled to claim)?*

All tiny upstream decisions crystallize into a statement about the world.

1. **Scope Creep**
2. **Causal language from associated evidence**
3. **Certainty miscalibration**

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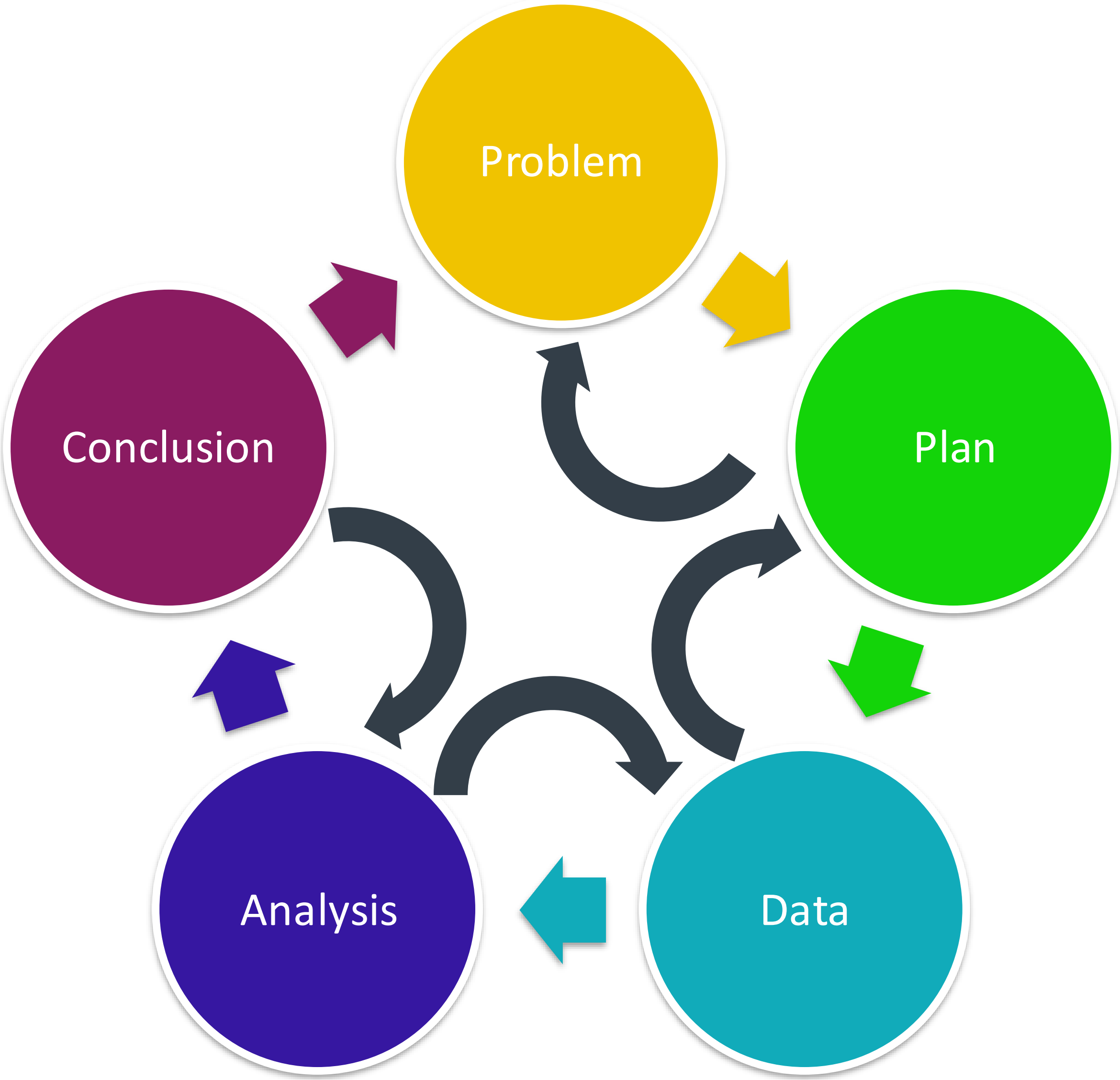
PPDAC – Iterative, not linear

- Ability to interrogate your data
- Each step can re-visit any other step



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Simpson’s Paradox

Activity 1

Simpson's Paradox Example

Which treatment is more effective?

Treatment	Successful Outcomes	Total Procedures	Overall Success Rate
A	273	350	78%
B	289	350	83%

Simpson’s Paradox Example

Which treatment do you choose?

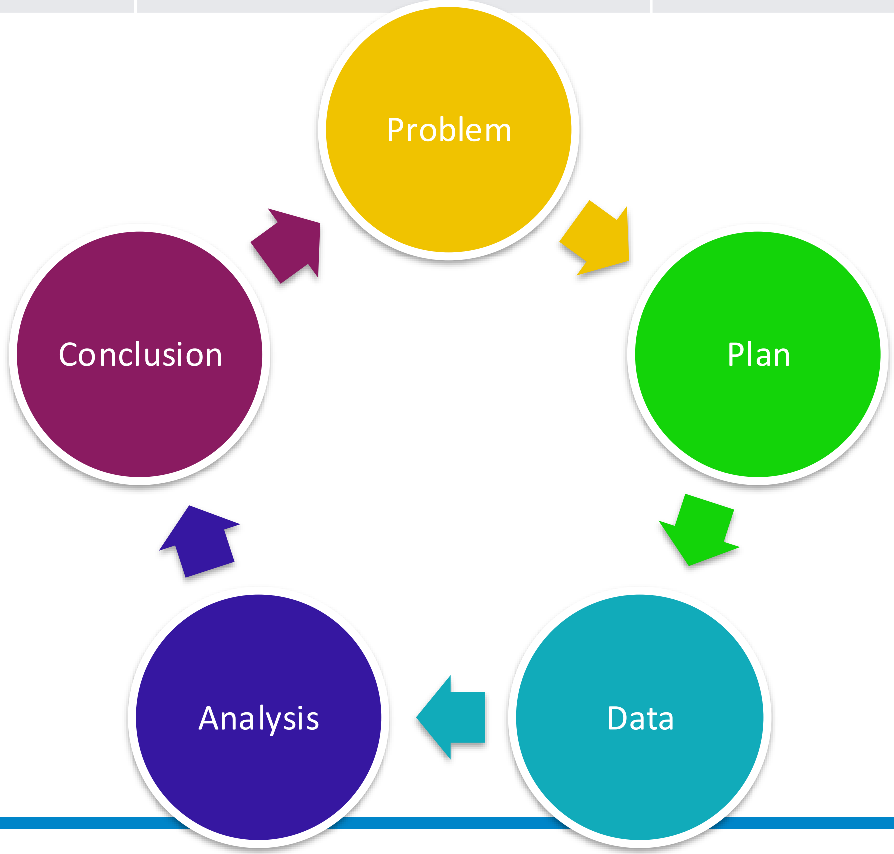
Treatment	Successful Outcomes	Total Procedures	Overall Success Rate
A	273	350	78%
B	289	350	83%

Stratified data

Severity	Treatment A success %	Treatment B success %
Mild	81/87	234/270
Severe	192/263	55/80

Which treatment is more effective?

Treatment	Successful Outcomes	Total Procedures	Overall Success Rate
A	273	350	78%
B	289	350	83%



The 4 frameworks as one lens

Lecture

	Kahneman	Polya	Paul-Elder	Wild &Pfannkuch
Define Problem	Fast thinking(system 1) defines the question	Understand What is known and unknown Draw a diagram	Purpose + Question at issue why is this claim being made?	Problem Real-world versus statistical question
Failure: we ask the wrong question (quickly, confidently and without noticing!)				
Plan the approach	Slow down: engage system 2	Devise a plan Break down strategy into steps	Assumptions + concepts	Plan Study design
Failure: hidden assumptions set the ceiling on conclusions before any work begins				
Execute the work	Beware of sneaky system 1 replacing hard question with easy question	Execute plan Carry out strategy Check each step Does this make sense?	Information + inference What does the evidence say?	Data + Analysis Is what we collected what we planned?
Failure: we collect evidence that confirms what we already believe				
Review conclusion	Calibrate; pre-mortem	Look back Can you solve it differently?	Implications + POV Scope of claim	Conclusion What can we actually claim?

Four Frameworks – additional tools for Solving Problems

Other cognitive forcing tools:

1. Mind mapping/ concept flowcharts
   - Externalizes structure of the problem
2. Checklists
   - Converts implicit expert knowledge into auditable steps
3. Rubber duck thinking
   - Stating logic and assumptions out loud
4. Ethical considerations
   - Stakeholder harm

The Turing Audit

Activity #3

The Turing Audit - AI-Generated Literature Summary

The Impact of AI-Assisted Diagnostic Tools on Clinical Decision Accuracy in Emergency Medicine

Recent advances in large language models (LLMs) and clinical decision support systems suggest that AI-assisted diagnostic tools significantly improve physician accuracy and reduce diagnostic delays in emergency medicine settings. Across multiple studies published between 2021 and 2025, AI systems consistently outperformed unaided clinicians in identifying rare conditions, prioritizing differential diagnoses, and recommending appropriate imaging and laboratory testing.

A meta-analysis by Chen et al. (2024) reported that AI-assisted diagnostic workflows improved overall diagnostic accuracy by approximately 18% relative to standard physician-only workflows. Importantly, the benefits appeared strongest among less experienced clinicians, suggesting that AI systems may reduce expertise-related disparities in care quality.

Several observational studies also demonstrated reductions in patient wait times and emergency department overcrowding after implementation of AI triage systems. For example, a multicenter study involving six urban hospitals found that AI-supported triage reduced average emergency department length of stay by nearly 25%, while simultaneously increasing diagnostic concordance with final discharge diagnoses.

Researchers have argued that these systems improve care primarily by reducing cognitive overload and minimizing anchoring bias during high-volume clinical periods. In one randomized simulation study, physicians using AI support generated broader and more complete differential diagnoses than control participants. These findings align with cognitive science literature suggesting that structured external reasoning aids can improve complex decision-making under uncertainty.

Although some studies noted occasional hallucinated recommendations or fabricated citations produced by generative AI systems, these errors were generally infrequent and were readily identified by supervising clinicians. Overall, the literature suggests that risks associated with AI deployment are manageable within existing clinical oversight frameworks.

Notably, most published studies report positive clinician attitudes toward AI integration. Surveys of emergency medicine physicians indicate that clinicians increasingly view AI as a collaborative partner rather than a replacement technology. This growing trust may facilitate future implementation efforts and improve adoption across hospital systems.

Taken together, current evidence strongly supports broader deployment of AI-assisted diagnostic systems in emergency medicine. Given ongoing clinician shortages and rising patient volumes, delaying implementation may itself introduce preventable harms through missed diagnoses and workflow inefficiencies.

Assume this summary was generated for a hospital committee making procurement decisions. Your task is to audit the reasoning quality before institutional adoption.

The Turing Audit - AI-Generated Literature Summary

20 minutes:

- Group 1 —> Polya lens:
  - Did the AI correctly understand the problem, or did it subtly reframe it into an easier one? Can you reverse-engineer its plan? Is the conclusion consistent with the evidence cited?
- Group 2 —> Paul-Elder lens:
  - Identify logic gaps, hidden assumptions, and unstated points of view. Use the Day 2 intellectual standards rubric (clarity, accuracy, relevance, scored 1–3).
- Group 3 —> Wild/Pfannkuch lens:
  - Are there statistical traps? Base rate neglect? Aggregation without stratification? Conclusions that outrun the data?

Present your top two (or three) findings

Pre-Mortem

Activity

Pre-Mortem

PredictAI: An AI-Assisted Early Detection System for Sepsis in Emergency Departments

Project Overview

Sepsis remains a leading cause of preventable mortality in hospital emergency departments (EDs), in part because early symptoms are often nonspecific and difficult to distinguish from less severe conditions. Delayed recognition contributes substantially to morbidity, mortality, and healthcare costs. This project proposes development and pilot implementation of an AI-assisted clinical decision support system ("PredictAI") designed to identify patients at elevated risk for sepsis during the first two hours of emergency department admission. The interdisciplinary project team includes: emergency medicine physicians, biomedical informaticians, statisticians, machine learning researchers, and health systems administrators.

The proposed system will integrate: electronic health record (EHR) data, laboratory values, triage notes, vital signs, and clinician-entered symptom descriptions.

A large language model (LLM) component will summarize patient histories and generate risk explanations for clinicians. A machine learning classifier trained on retrospective hospital data will produce a real-time sepsis risk score.

Pre-Mortem

Aim 1

- Develop a multimodal machine learning model capable of predicting sepsis onset within 6 hours of ED admission.

Aim 2

- Integrate an LLM-based clinical summarization assistant into physician workflows to improve interpretability and reduce cognitive burden during triage.

Aim 3

- Evaluate whether deployment of PredictAI reduces:
- time-to-antibiotic administration,
- ICU admissions,
- and sepsis-related mortality.

Pre-mortem

Proposed Study Design

The project will use retrospective EHR data from three urban academic medical centers collected between 2019–2024 to train the predictive model. The pilot implementation phase will occur in one partner hospital over a 6-month period.

<u>Primary outcomes:</u>

reduction in average time-to-sepsis diagnosis, clinician satisfaction scores, overall diagnostic accuracy.

<u>Secondary outcomes:</u>

reduction in ED overcrowding, reduced clinician burnout, improved patient trust in AI-supported care.

Pre-mortem

Innovation Statement

Unlike existing sepsis screening tools, PredictAI combines:

- Deep learning prediction,
- generative AI summarization,
- and real-time workflow integration.

The investigators anticipate that the system will improve diagnostic consistency while simultaneously reducing physician cognitive overload during high-volume ED operations.

The project team argues that recent advances in generative AI make clinically deployable decision support systems feasible at scale for the first time.

Pre-mortem

Preliminary Evidence

Prior pilot studies from the literature suggest:

- AI-assisted triage systems can improve diagnostic accuracy by 10–20%,
- clinicians report generally positive attitudes toward AI integration,
- and machine learning systems may identify subtle deterioration patterns earlier than human clinicians alone.

The team has already achieved preliminary model performance with:

- Area Under the Curve (AUC): 0.91
- Sensitivity: 92%
- Specificity: 81%

Projected Impact

The investigators believe PredictAI could serve as a scalable model for future AI-assisted emergency medicine systems nationwide. If successful, the platform may reduce preventable sepsis deaths while helping hospitals address increasing clinician shortages and patient volumes.

Pre-mortem

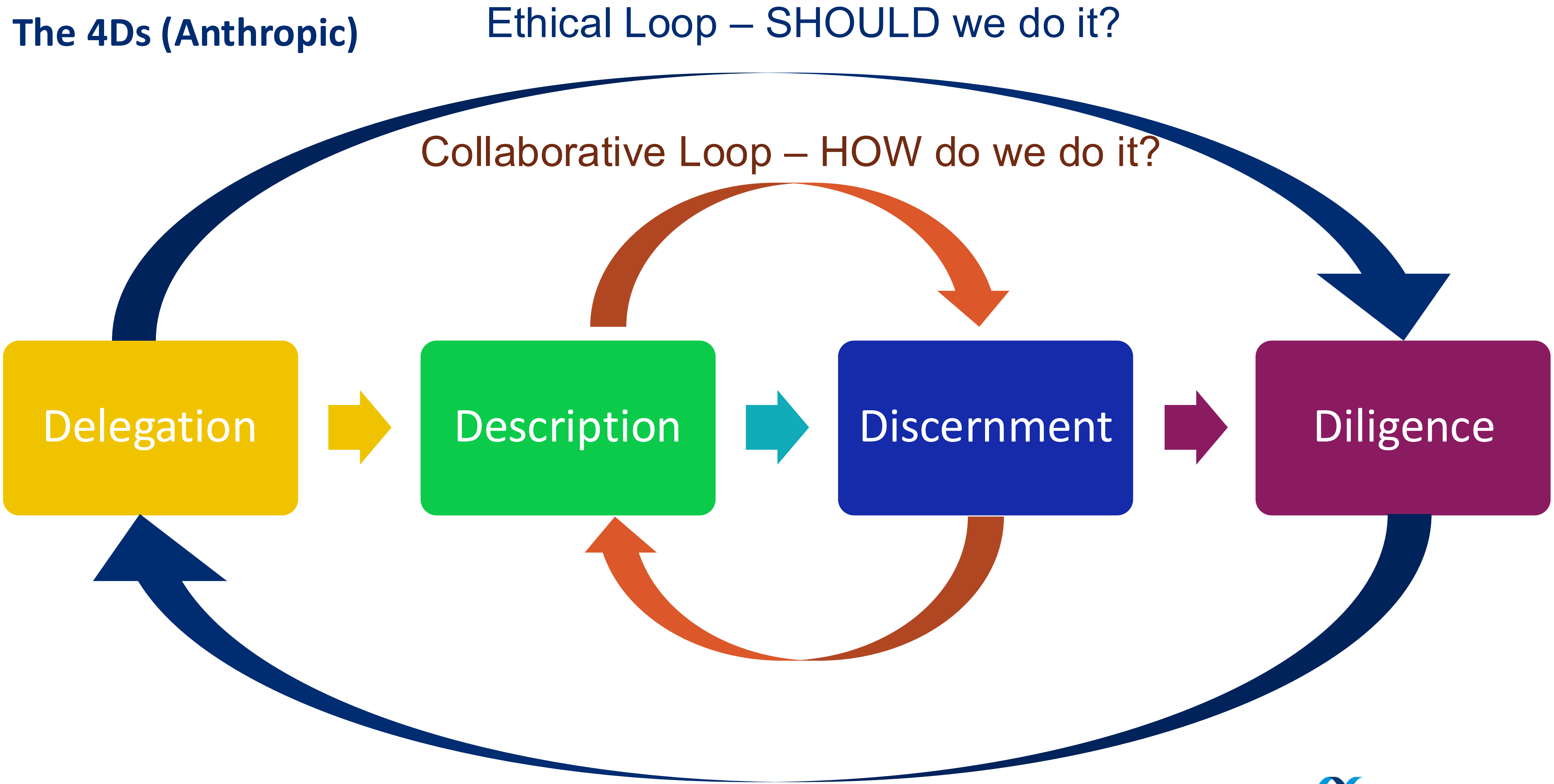
Phase	Timeline
Data integration and cleaning	Months 1–4
Model training and validation	Months 5–8
LLM workflow integration	Months 7–9
Pilot deployment	Months 10–15
Outcome analysis	Months 16–18

Personal Heuristic Protocol

Delegation Mapping

Should I use Ai here? Discussion

The 4Ds (Anthropic)



Delegation mapping

		Human Prompter Domain Expertise	
		Low expertise	High expertise
Outcome	High stakes	CAUTION ZONE • Human oversight essential	• **Human Judgment first** • Ai in support role
	Low stakes	Strong delegation candidate • Ai completes, • Human verifies	Strong delegation candidate • Ai drafts, • Human discernment

The Three Questions (from Day 1)

1. What do I know?
2. What don’t I know?
3. How much does that ignorance matter?

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Wild & Pfannkuch

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No single framework is sufficient. Each one reveals a different failure layer

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Solving Problems: Day 3 – Fill out form